

Timilehin Adedayo

ML / AI Engineer

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SUMMARY

AI/ML Engineer specializing in building and deploying production machine learning systems across healthcare, fintech, and agriculture. Skilled in end-to-end ML pipeline architecture, RAG systems, computer vision, and model optimization. Experienced leading technical development on BRAIN, a medical imaging AI system for breast cancer diagnosis.

EDUCATION

Babcock University

Ogun, Nigeria

B.Sc. Computer Science

Graduation: July 2026 | CGPA: 4.05/5

Coursework: Machine Learning, Computer Vision, Statistics, Data Structures & Algorithms.

EXPERIENCE

AI Consultant (Part-time) | Acturian

Feb 2026 – Present

Contracted to architect and deploy machine learning pipelines for automated credit risk scoring and fraud detection.

Stack: Python, FastAPI, Transformers.

- Built end-to-end ML pipelines scoring credit risk by synthesizing data from multiple structured and unstructured sources.
- Achieved an ~18% lift in fraud classification accuracy by fine-tuning transformer models on a proprietary dataset of over 50,000 transactions, outperforming the previous baseline.
- Designed and shipped FastAPI microservices that deliver inference results in sub-100ms, supporting real-time internal underwriting decisions.

AI Team Lead | CortexAI

2025 – Present

Lead the development of resource-efficient, retrieval-augmented computer vision systems for healthcare applications.

Stack: PyTorch, Swin Transformer, React.

- Spearheading development of BRAIN, a diagnostic framework for breast cancer detection using mammography.
- Architected a hybrid system combining a Swin Transformer vision model with a clinical knowledge base to improve diagnostic precision.
- Streamlined model architecture for edge deployment, reducing parameter count by ~30% to enable high-performance execution on resource-constrained devices.

Software Developer Intern | International Institute of Tropical Agriculture (IITA)

Jan – Jun 2025

Stack: Python, TensorFlow, CNNs.

- Trained and enhanced CNN models to detect cassava disease from field-captured imagery, achieving high validation accuracy (~94%) on a diverse dataset.
- Managed the labeling and preprocessing of over 10,000 maize crop images in collaboration with agronomists to build robust pest-detection models.

ML Engineering Intern | Meridian Analytics

May – Aug 2024

Stack: Python, LSTM, Prophet, Isolation Forest.

- Created time-series forecasting models using LSTM and Prophet to predict energy consumption across multiple commercial property portfolios.
- Implemented an early-warning failure detection system using Isolation Forest and autoencoders, identifying potential equipment issues with a meaningful improvement in lead time.

SELECTED PROJECTS

BRAIN | Breast Retrieval-Augmented Intelligence Network | CortexAI Oct 2025

Stack: Python, PyTorch, FAISS, FastAPI, React, Grad-CAM++.

- Designed an explainable AI system for breast cancer detection using a Swin-B backbone, achieving 98.99% accuracy on a curated test set of nearly 5,000 mammography images.
- Built a RAG verification module using FAISS to cross-reference model predictions against a large clinical knowledge base (~24,000 images), enhancing diagnostic confidence.
- Implemented visual attribution maps (Grad-CAM++) to offer transparent decision support for radiologists.

gdg-bu-kg | Enterprise RAG Platform for Academic Data | GDG Babcock University March 2026

Stack: Neo4j Aura, FastAPI, React, Pydantic v2, SQLAlchemy 2.0, PostgreSQL.

- Architected a high-precision RAG platform integrating LLMs with a Neo4j Knowledge Graph containing over 10,000 academic entities.
- Authored and published a Python SDK that improved query handling, resulting in a significant reduction in end-to-end latency for academic data retrieval.

Yami | P2P Student Lending Platform | Hackathon Finalist, Paystack-Sponsored Ideathon Feb 2026

- Collaborated in a team build at a Paystack-sponsored ideathon to develop a P2P student lending platform.
- Created a Python/FastAPI AI microservice specifically for repayment prediction and adaptive loan structuring.
- Designed a behavioral trust scoring model that served as the core credit infrastructure layer for the application.

Dolphin LLM | Memory-Efficient AI Alignment Research | Independent Research Oct 2025

Stack: Python, PyTorch, Hugging Face (PEFT, Transformers, Accelerate).

- Fine-tuned a 7B parameter model using parameter-efficient fine-tuning (LoRA) on a custom alignment dataset to evaluate safety guardrail degradation.
- Demonstrated that prioritizing honesty over harmlessness reduced unwarranted refusal rates for benign boundary-case prompts.
- Documented findings on instruction-following, honesty, and directness trade-offs as part of independent AI safety coursework.

LEADERSHIP & VOLUNTEERING

Project Manager | GDG Babcock University Sep 2025 – May 2026

- Lead the developer team in executing complex AI projects, including the University Knowledge Graph Ecosystem.
- Responsible for project architecture, roadmap planning, and cross-functional team coordination.

Data & AI Track Lead | GDG Babcock University Sep 2025 – May 2026

- Organise and host technical training sessions for ML careers, engaging 100+ students and mentoring upcoming AI developers through workshops and hands-on sessions.

AI Technical Mentor | Babcock Tech Hub Sep 2025 – May 2026

- Served as the primary AI technical mentor, guiding students through hands-on machine learning projects covering computer vision, NLP, and model deployment.
- Conducted weekly office hours and code reviews for mentees building their first end-to-end ML pipelines.

TECHNICAL SKILLS

ML & AI:	Computer Vision (CNNs, Vision Transformers), RAG Architecture, LLM Fine-tuning (LoRA/PEFT), Agentic AI (LangChain, Tool Calling), NLP (BERT, NER, Sentiment Analysis), Time-Series Forecasting (LSTM, Prophet), Anomaly & Fraud Detection (Autoencoders, GANs), Explainable AI (Grad-CAM++, SHAP), Knowledge Graphs.
Languages & Frameworks:	Python (PyTorch, TensorFlow, FastAPI, Scikit-Learn, Pydantic), JavaScript (React, Node.js), SQL.
Infrastructure & Tools:	Neo4j Aura, PostgreSQL, Docker, FAISS, Hugging Face, GitHub.